先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 11 – Dimensionality Reduction: Linear, Global, and Local Structure



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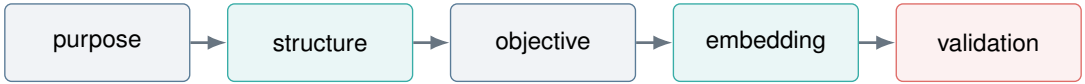
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The preservation question comes first



Choose the invariant before choosing the algorithm.

Guiding question

What must remain trustworthy after mapping $x \in \mathbb{R}^d$ to $z \in \mathbb{R}^m$?

- variance along important directions
- global pairwise distances
- local neighborhoods
- performance of a downstream task

Learning outcomes for three periods

By the end

You can derive PCA, distinguish global and local objectives, diagnose distorted embeddings, and build a leakage-safe reduction pipeline.

Period	Decision	Methods
1	Which linear subspace?	PCA and SVD
2	Which global relation?	MDS and kernel PCA
3	Which neighborhood?	Isomap, LLE, t-SNE, UMAP

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A map is useful only relative to a task

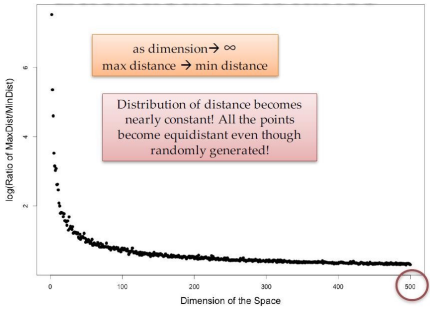
preserve a quantity	optimize a surrogate	audit the distortion

What to calculate

Every method in this lecture will be read through four questions: what is preserved, what is optimized, what can fail, and how failure is measured.

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Why do familiar distances lose contrast as dimension grows?



Question

Why do familiar distances lose contrast as dimension grows?

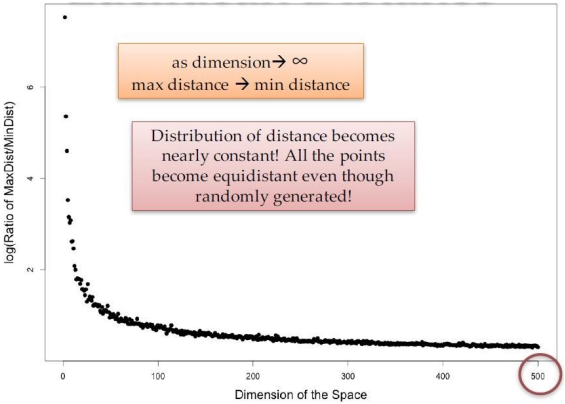
Example

a 500-feature sensor signature

Main idea

Sparse coverage makes nearest and farthest neighbors increasingly similar in relative distance.

Distances become less distinct in high dimensions



Distance concentration depends on the sampling distribution and distance measure.

High-dimensional geometry: model and variables

Model and assumptions

High-dimensional observations occupy a space whose volume grows much faster than a fixed sample can cover.

$$\text{contrast} = \frac{d_{\max} - d_{\min}}{d_{\min}}$$

Symbol	Meaning	Status
d	ambient dimension	given by the representation
$d_{\min}$	nearest distance	depends on metric and scaling
$d_{\max}$	farthest distance	depends on sample support

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High-dimensional geometry: worked example

Quantity	Value	Meaning
dimension	10	500
nearest/farthest ratio	0.42	0.91
interpretation	contrast remains	contrast collapses

What the result means

If neighbor identities change under small perturbations, do not treat the original metric as a reliable local geometry.

High-dimensional geometry: what to check

Main idea

Sparse coverage makes nearest and farthest neighbors increasingly similar in relative distance.

What to check

neighbor overlap under resampling

Limit

distance concentration can make a nearest-neighbor graph unstable

High-dimensional geometry: implementation

1. inspect feature scales
2. compare distance distributions
3. repeat neighbor search under perturbation
4. reduce only after defining the task

Check the result

Check neighbor overlap under resampling together with the fitted transformation and data split.

Common mistake

distance concentration can make a nearest-neighbor graph unstable

High-dimensional geometry: common mistakes

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Example

a 500-feature sensor signature

Explain the calculation and check neighbor overlap under resampling, using a small example.

Common mistake

distance concentration can make a nearest-neighbor graph unstable

What to do

If neighbor identities change under small perturbations, do not treat the original metric as a reliable local geometry.

High-dimensional geometry: check your understanding

Question

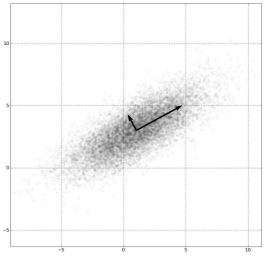
Why do familiar distances lose contrast as dimension grows?

Answer

Sparse coverage makes nearest and farthest neighbors increasingly similar in relative distance.

Decision. If neighbor identities change under small perturbations, do not treat the original metric as a reliable local geometry.

Which variation should PCA be allowed to call important?



Question

Which variation should PCA be allowed to call important?

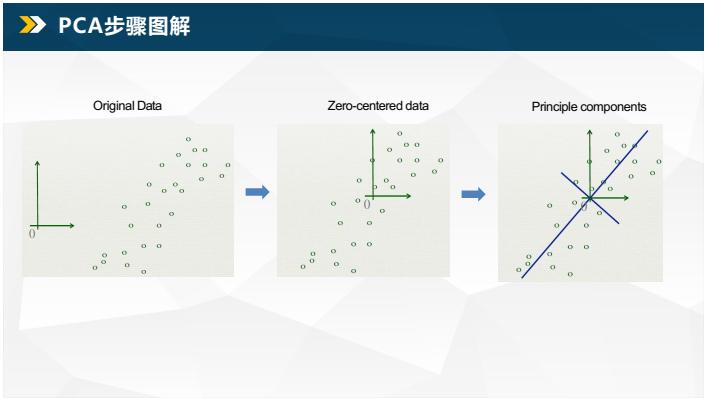
Example

mixed-unit industrial measurements

Main idea

Centering changes the origin, while scaling changes the geometry and therefore the principal directions.

Center data before finding principal directions



Centering, scaling, and covariance: model and variables

Model and assumptions

PCA describes variation around a center; its axes depend on the units used for each feature.

$$S = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})(x_i - \bar{x})^\top$$

Symbol	Meaning	Status
$\bar{x}$	training mean	fit once on training data
S	covariance matrix	symmetric positive semidefinite
units	feature scale	a modeling choice

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Centering, scaling, and covariance: worked example

Quantity	Value	Meaning
feature	temperature	pressure
raw variance	4	10,000
after z-score	1	1

What the result means

Use covariance PCA when physical variance is meaningful; use correlation PCA when unit differences are arbitrary.

The choice rule for Centering, scaling, and covariance

Main idea

Centering changes the origin, while scaling changes the geometry and therefore the principal directions.

What to check

covariance condition number

Limit

a large-unit nuisance feature can dominate the first component

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Centering, scaling, and covariance: implementation

1. split data first
2. fit mean and scale on training rows
3. transform validation rows with stored values
4. record the units used

Check the result

Check covariance condition number together with the fitted transformation and data split.

Common mistake

a large-unit nuisance feature can dominate the first component

Centering, scaling, and covariance: common mistakes

Example

mixed-unit industrial measurements

Explain the calculation and check covariance condition number, using a small example.

Common mistake

a large-unit nuisance feature can dominate the first component

What to do

Use covariance PCA when physical variance is meaningful; use correlation PCA when unit differences are arbitrary.

Centering, scaling, and covariance: check your understanding

Question

Which variation should PCA be allowed to call important?

Answer

Centering changes the origin, while scaling changes the geometry and therefore the principal directions.

Decision. Use covariance PCA when physical variance is meaningful; use correlation PCA when unit differences are arbitrary.

A projection measures position along a direction

内积和投影

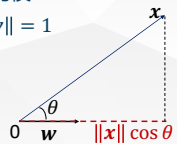
■两个向量x和w内积：

$$\boldsymbol{w}^{\mathrm{T}}\boldsymbol{x} = (w_1 \quad w_2 \quad \cdots \quad w_D)\begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_D \end{pmatrix}$$

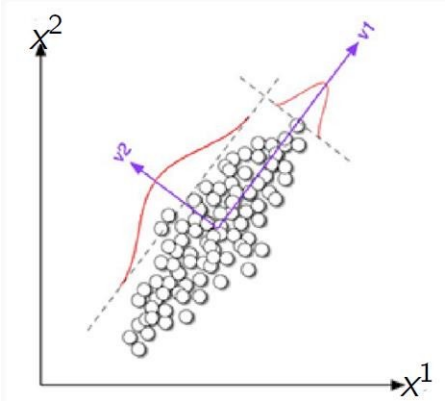
■从几何角度，内积等于x到w的投影长度乘以w的模

- 只关心方向，令模长为1：w为单位向量，$\|\boldsymbol{w}\| = 1$

$$\boldsymbol{x} \cdot \boldsymbol{w} = \|\boldsymbol{x}\|\|\boldsymbol{w}\|\cos\theta = \|\boldsymbol{x}\|\cos\theta$$



PCA maximizes projected variance



Choose the unit direction with the largest projected variance.

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PCA as maximum variance: model and variables

Model and assumptions

The first principal component is the unit vector whose projected scores have maximum variance.

$$w_1 = \arg\max_{\|w\|=1} w^\top S w, \qquad S w_1 = \lambda_1 w_1$$

Symbol	Meaning	Status
w_1	first loading	unit direction
$w_1^\top x_i$	score	coordinate on the component
λ_1	captured variance	largest eigenvalue

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PCA as maximum variance: worked example

Quantity	Value	Meaning
covariance	$\mathrm{diag}(9, 1)$	
best direction	$(1, 0)^\top$	
captured variance	9	90% of total

What the result means

Use PCA for an unsupervised variance objective, not as evidence that the retained axes are label-relevant.

PCA as maximum variance: checking the result

Main idea

The Rayleigh quotient turns variance maximization into an eigenvalue problem.

What to check

explained variance ratio

Limit

maximum variance can preserve nuisance variation rather than predictive signal

What to record when running PCA as maximum variance

```
from sklearn.decomposition import PCA
pca = PCA(n_components=2, svd_solver='full')
Z_pca = pca.fit_transform(X)
print('PCA variance ratios:', pca.explained_variance_ratio_.round(3))
```


Evidence and failure check

Check explained variance ratio with the fitted transformation and data split; reject the run if maximum variance can preserve nuisance variation rather than predictive signal.

The warning sign for PCA as maximum variance

Example

two correlated imaging measurements

Explain the calculation and check explained variance ratio, using a small example.

Common mistake

maximum variance can preserve nuisance variation rather than predictive signal

What to do

Use PCA for an unsupervised variance objective, not as evidence that the retained axes are label-relevant.

Explain PCA as maximum variance without naming the algorithm

Question

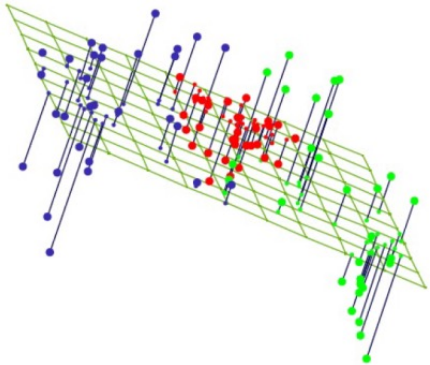
Which unit direction captures the largest projected spread?

Answer

The Rayleigh quotient turns variance maximization into an eigenvalue problem.

Decision. Use PCA for an unsupervised variance objective, not as evidence that the retained axes are label-relevant.

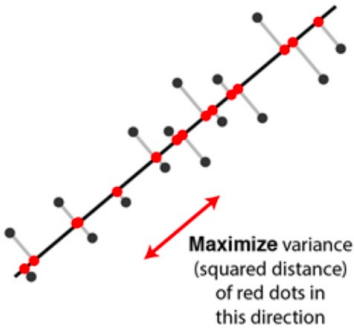
PCA minimizes squared reconstruction error

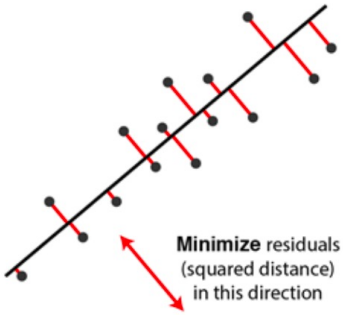


Reconstruction error is the squared distance from each original point to its projection.

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PCA links projected variance and reconstruction error





For centered data and orthogonal projection, maximizing retained variance minimizes squared reconstruction error.

How PCA as minimum reconstruction error changes the representation

Model and assumptions

An orthogonal rank-m projection reconstructs each centered point from its coordinates inside a subspace.

$$\min_{W^\top W = I} \sum_i \|x_i - WW^\top x_i\|_2^2$$

Symbol	Meaning	Status
W	retained basis	$d \times m$ with orthonormal columns
$WW^\top x_i$	reconstruction	orthogonal projection
residual	discarded component	perpendicular to the subspace

Check the direction of change in PCA as minimum reconstruction error

Quantity	Value	Meaning
total variance	12	
retained	9	
residual	3	25% of total

What the result means

Audit reconstruction by subgroup and feature before using compression in a safety-sensitive pipeline.

When PCA as minimum reconstruction error earns a place in the pipeline

Main idea

Total centered energy splits into retained variance plus orthogonal residual energy.

What to check

mean squared reconstruction error

Limit

small average error can hide a large error for a rare subgroup

PCA as minimum reconstruction error: implementation

1. fit PCA on centered training data
2. transform to scores
3. inverse-transform scores
4. plot residuals by subgroup

Check the result

Check mean squared reconstruction error together with the fitted transformation and data split.

Common mistake

small average error can hide a large error for a rare subgroup

When the objective of PCA as minimum reconstruction error is not enough

Example

compressing correlated waveform channels

Explain the calculation and check mean squared reconstruction error, using a small example.

Common mistake

small average error can hide a large error for a rare subgroup

What to do

Audit reconstruction by subgroup and feature before using compression in a safety-sensitive pipeline.

A final decision on PCA as minimum reconstruction error

Question

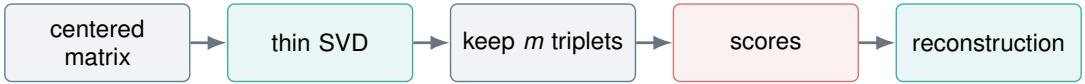
Why does the variance solution also minimize squared reconstruction error?

Answer

Total centered energy splits into retained variance plus orthogonal residual energy.

Decision. Audit reconstruction by subgroup and feature before using compression in a safety-sensitive pipeline.

How can PCA be computed without explicitly forming covariance?



Thin SVD is stable and efficient when the data matrix is rectangular or sparse.

Question

How can PCA be computed without explicitly forming covariance?

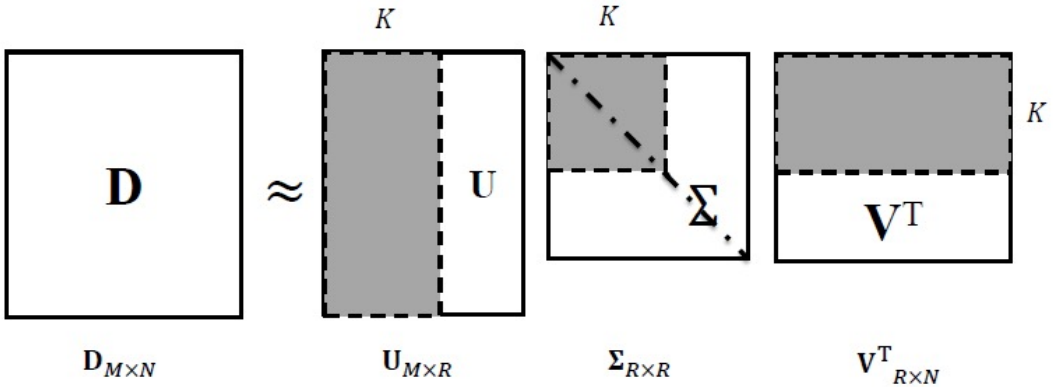
Example

a tall gene-expression matrix

Main idea

Thin SVD is stable and efficient when the data matrix is rectangular or sparse.

SVD gives principal directions and component strengths



For centered $X = U\Sigma V^\top$, columns of V are principal directions.

The variables that make SVD, rank, and component count work

Model and assumptions

The singular value decomposition exposes principal directions, scores, and explained variance in one factorization.

$$X_c = U\Sigma V^\top, \qquad Z_m = U_m\Sigma_m = X_c V_m$$

Symbol	Meaning	Status
V_m	loadings	right singular vectors
$U_m\Sigma_m$	scores	low-dimensional coordinates
σ_j^2/n	variance	eigenvalue of covariance

Read the numbers produced by SVD, rank, and component count

Quantity	Value	Meaning
singular values	$8, 4, 1$	
variance energy	$64, 16, 1$	
first two	$80/81$	98.8%

What the result means

Choose m from the use case, then report sensitivity across nearby values.

Three conditions for trusting SVD, rank, and component count

Main idea

Thin SVD is stable and efficient when the data matrix is rectangular or sparse.

What to check

cumulative explained variance

Limit

a fixed variance threshold may retain too many noisy components or discard a weak predictive direction

An implementation checklist for SVD, rank, and component count

1. plot cumulative variance
2. check reconstruction
3. cross-validate downstream performance
4. prefer randomized SVD only when scale requires it

Check the result

Check cumulative explained variance together with the fitted transformation and data split.

Common mistake

a fixed variance threshold may retain too many noisy components or discard a weak predictive direction

Reject SVD, rank, and component count under this condition

Example

a tall gene-expression matrix

Explain the calculation and check cumulative explained variance, using a small example.

Common mistake

a fixed variance threshold may retain too many noisy components or discard a weak predictive direction

What to do

Choose m from the use case, then report sensitivity across nearby values.

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SVD, rank, and component count: check your understanding

Question

How can PCA be computed without explicitly forming covariance?

Answer

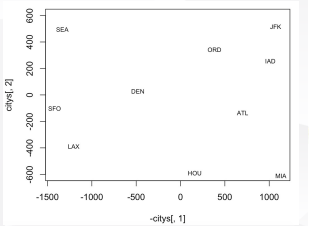
Thin SVD is stable and efficient when the data matrix is rectangular or sparse.

Decision. Choose m from the use case, then report sensitivity across nearby values.

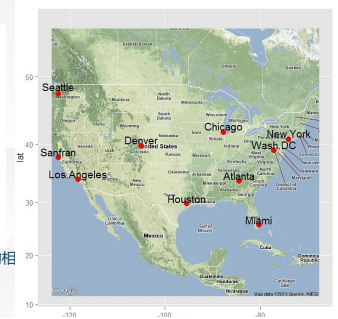
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MDS builds a map from pairwise distances

例：绘制地图（城市嵌入）



在平面坐标上据此标出这10个城市之间的相对位置，使之尽可能接近表中的距离数据



J	K
	IAD
2182	543
1737	597
1021	1494
1891	1220
959	2300
2734	923
2408	205
678	2442
0	2329
2329	0

Compare distances before and after MDS

例子：MDS

2. 计算中心化矩阵

$$\boldsymbol{J} = \boldsymbol{I}_N - \frac{1}{N}\mathbf{1}\mathbf{1}^{\mathrm{T}} = \begin{pmatrix} 0.8 & -0.2 & -0.2 & -0.2 & -0.2 \\ -0.2 & 0.8 & -0.2 & -0.2 & -0.2 \\ -0.2 & -0.2 & 0.8 & -0.2 & -0.2 \\ -0.2 & -0.2 & -0.2 & 0.8 & -0.2 \\ -0.2 & -0.2 & -0.2 & -0.2 & 0.8 \end{pmatrix}$$

3. 计算距离平方矩阵d_{ij}^2

$$\boldsymbol{D} = \begin{pmatrix} 0.00 & 21.98 & 13.99 & 6.00 & 11.02 \\ 21.98 & 0.00 & 43.03 & 11.02 & 43.03 \\ 13.99 & 43.03 & 0.00 & 28.00 & 6.00 \\ 6.00 & 11.02 & 28.00 & 0.00 & 28.00 \\ 11.02 & 43.03 & 6.00 & 28.00 & 0.00 \end{pmatrix}$$

4. 用中心化矩阵$\boldsymbol{J}$和距离平方矩阵$\boldsymbol{D}$计算内积矩阵$\boldsymbol{B}$

$$\boldsymbol{B} = -\frac{1}{2}\boldsymbol{J}\boldsymbol{D}\boldsymbol{J} = \begin{pmatrix} 2.12 & -2.27 & -1.07 & 1.12 & 0.11 \\ -2.27 & 15.33 & -8.99 & 5.21 & -9.29 \\ -1.07 & -8.99 & 9.72 & -6.07 & 6.42 \\ 1.12 & 5.21 & -6.07 & 6.12 & -6.37 \\ 0.11 & -9.29 & 6.42 & -6.37 & 9.13 \end{pmatrix}$$

5：计算$\boldsymbol{B}$的特征值和特征向量

$\lambda_1 \approx 32.32$, $\boldsymbol{v}_1 \approx (-0.02 \quad 0.63 \quad -0.48 \quad 0.36 \quad -0.49)$

$\lambda_2 \approx 6.39$, $\boldsymbol{v}_2 \approx (-0.53 \quad 0.59 \quad 0.33 \quad -0.50 \quad 0.10)$

Classical multidimensional scaling: model and variables

Model and assumptions

Classical MDS finds Euclidean coordinates whose inner products reproduce double-centered squared distances.

$$B = -\frac{1}{2} J D^{(2)} J, \qquad J = I - \frac{1}{n} \mathbf{1}\mathbf{1}^\top$$

Symbol	Meaning	Status
$D^{(2)}$	squared distances	pairwise input
J	centering operator	removes translation
B	Gram matrix	estimated inner products

Classical multidimensional scaling: worked example

Quantity	Value	Meaning
points	$0, 1, 3$	
distances	$1, 2, 3$	
embedding	one dimension	exact up to rotation

What the result means

Treat large negative eigenvalues as evidence that the requested Euclidean map is incompatible with the distances.

A decision card for Classical multidimensional scaling

Main idea

Distance geometry becomes an eigendecomposition after double centering.

What to check

stress and residual distance plot

Limit

non-Euclidean dissimilarities can produce negative eigenvalues and misleading coordinates

Reproduce the result of Classical multidimensional scaling

1. verify distance symmetry
2. form squared-distance matrix
3. inspect negative eigenvalues
4. retain positive leading eigenpairs

Check the result

Check stress and residual distance plot together with the fitted transformation and data split.

Common mistake

non-Euclidean dissimilarities can produce negative eigenvalues and misleading coordinates

A stress test for Classical multidimensional scaling

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Example

mapping cities from travel-time dissimilarities

Explain the calculation and check stress and residual distance plot, using a small example.

Common mistake

non-Euclidean dissimilarities can produce negative eigenvalues and misleading coordinates

What to do

Treat large negative eigenvalues as evidence that the requested Euclidean map is incompatible with the distances.

Classical multidimensional scaling: check your understanding

Question

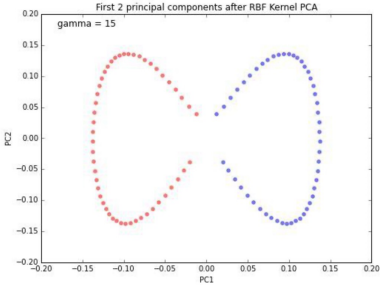
Can coordinates be recovered from a matrix of pairwise distances?

Answer

Distance geometry becomes an eigendecomposition after double centering.

Decision. Treat large negative eigenvalues as evidence that the requested Euclidean map is incompatible with the distances.

How can a linear PCA calculation reveal a nonlinear coordinate system?



Question

How can a linear PCA calculation reveal a nonlinear coordinate system?

Example

separating concentric nonlinear patterns

Main idea

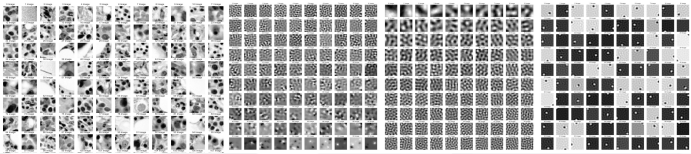
The kernel selects the similarity geometry; centering must occur in feature space.

Kernel PCA changes the representation

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例：

- 输入1000幅28*28图像，分别采用KPCA、PCA和ICA降维，得到的基向量/投影方向



训练图像　KPCA：先全局变化后局部变化　PCA：全局变化　ICA：局部变化

图片来自：https://towardsdatascience.com/kernel-pca-vs-pca-vs-ica-in-tensorflow-sklearn-60e17eb15a64

Kernel PCA: model and variables

Model and assumptions

Kernel PCA performs PCA in a feature space accessed only through centered kernel evaluations.

$$K_c = HKH, \qquad K_c\alpha_j = n\lambda_j\alpha_j$$

Symbol	Meaning	Status
K_{ij}	kernel similarity	$k(x_i, x_j)$
H	centering matrix	centers feature-space data
α_j	dual loading	weights training samples

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Kernel PCA: worked example

Quantity	Value	Meaning
kernel	RBF	
small γ	nearly linear	smooth geometry
large γ	near-identity	fragmented geometry

What the result means

Report the kernel and its scale; a kernel-PCA plot without them is not reproducible.

Kernel PCA: checking the result

Main idea

The kernel selects the similarity geometry; centering must occur in feature space.

What to check

kernel reconstruction surrogate

Limit

an extreme kernel width can collapse or fragment the representation

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Kernel PCA: implementation

1. scale inputs
2. tune kernel parameters without labels unless the task allows labels
3. inspect eigenvalue decay
4. validate pre-images or downstream use

Check the result

Check kernel reconstruction surrogate together with the fitted transformation and data split.

Common mistake

an extreme kernel width can collapse or fragment the representation

What breaks first in Kernel PCA

Example

separating concentric nonlinear patterns

Explain the calculation and check kernel reconstruction surrogate, using a small example.

Common mistake

an extreme kernel width can collapse or fragment the representation

What to do

Report the kernel and its scale; a kernel-PCA plot without them is not reproducible.

Kernel PCA: check your understanding

Question

How can a linear PCA calculation reveal a nonlinear coordinate system?

Answer

The kernel selects the similarity geometry; centering must occur in feature space.

Decision. Report the kernel and its scale; a kernel-PCA plot without them is not reproducible.

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When should shortest paths replace straight-line distances?

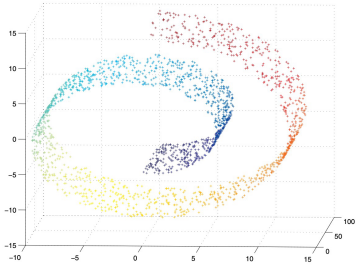


Figure 1: 2000 Random data points on the swiss roll.

Question

When should shortest paths replace straight-line distances?

Example

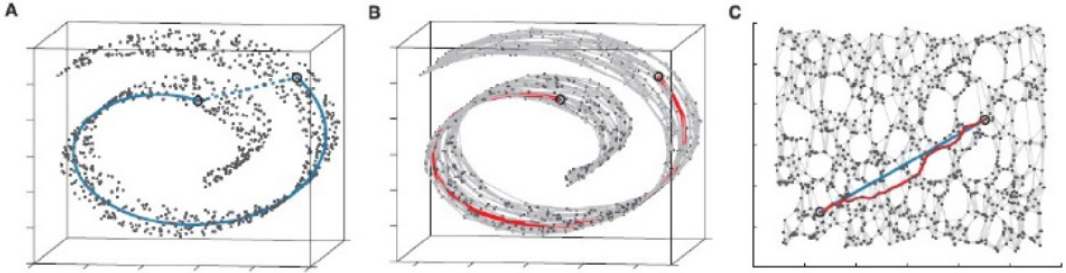
unrolling a Swiss-roll surface

Main idea

Local Euclidean edges approximate a curved manifold only when the graph is connected without shortcuts.

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Isomap follows paths along the manifold



Neighbor-graph paths approximate distances along the manifold.

Isomap and geodesic distance: model and variables

Model and assumptions

Isomap builds a neighborhood graph, estimates geodesic distances by shortest paths, then applies classical MDS.

$$d_G(i,j) = \min_{\pi: i \rightsquigarrow j} \sum_{(u,v)\in\pi} \|x_u - x_v\|_2$$

Symbol	Meaning	Status
G	neighbor graph	set by k or radius
π	graph path	sequence of local edges
d_G	geodesic estimate	shortest path length

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Isomap and geodesic distance: worked example

Quantity	Value	Meaning
too small k	disconnected	infinite paths
useful k	connected	follows manifold
too large k	shortcuts	flattened geometry

What the result means

Choose the smallest stable neighborhood that keeps the graph connected and avoids obvious shortcuts.

Accept or reject Isomap and geodesic distance

Main idea

Local Euclidean edges approximate a curved manifold only when the graph is connected without shortcuts.

What to check

residual variance of geodesic distances

Limit

short-circuit edges can destroy the manifold even when the output looks smooth

Isomap and geodesic distance: implementation

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```
from sklearn.manifold import Isomap
iso = Isomap(n_neighbors=10, n_components=2)
Z_iso = iso.fit_transform(X)
print('Isomap embedding:', Z_iso.shape)
```


Evidence and failure check

Check residual variance of geodesic distances with the fitted transformation and data split; reject the run if short-circuit edges can destroy the manifold even when the output looks smooth.

The hidden assumption in Isomap and geodesic distance

Example

unrolling a Swiss-roll surface

Explain the calculation and check residual variance of geodesic distances, using a small example.

Common mistake

short-circuit edges can destroy the manifold even when the output looks smooth

What to do

Choose the smallest stable neighborhood that keeps the graph connected and avoids obvious shortcuts.

Isomap and geodesic distance: check your understanding

Question

When should shortest paths replace straight-line distances?

Answer

Local Euclidean edges approximate a curved manifold only when the graph is connected without shortcuts.

Decision. Choose the smallest stable neighborhood that keeps the graph connected and avoids obvious shortcuts.

LLE uses local relations rather than all pairwise distances

- A curved manifold can be nearly flat inside a small neighborhood.
- Find neighbors in the original feature space.
- Reconstruct each point as a weighted combination of neighbors.
- The embedding aims to preserve these weights, not global Euclidean distances.

LLE preserves local reconstruction weights

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- For each point, find nearby points in the original space.
- Choose weights with $\sum_j w_{ij} = 1$ to minimize $\|x_i - \sum_j w_{ij} x_j\|^2$.
- Find low-dimensional coordinates minimizing $\sum_i \|z_i - \sum_j w_{ij} z_j\|^2$.
- Keep the weights fixed; constrain translation and scale to avoid a trivial solution.

Locally linear embedding: model and variables

Model and assumptions

LLE represents each point as a linear combination of its neighbors and preserves those weights in low dimension.

$$\min_W \sum_i \left\| x_i - \sum_{j \in \mathcal{N}_i} w_{ij} x_j \right\|^2, \quad \sum_j w_{ij} = 1$$

Symbol	Meaning	Status
$\mathcal{N}_i$	neighbors of i	chosen in input space
w_{ij}	reconstruction weight	zero outside the neighborhood
z_i	embedding coordinate	reuses the same weights

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Locally linear embedding: worked example

Quantity	Value	Meaning
local patch	three neighbors	
weights	$0.2, 0.3, 0.5$	sum to one
output	reuse weights	preserve local affine relation

What the result means

Use LLE only when local patches are plausibly linear and sampling density is adequate.

Separate a good fit from a good decision in Locally linear embedding

Main idea

The weights are invariant to local translation, rotation, and scaling under suitable neighborhoods.

What to check

trustworthiness and continuity

Limit

too few neighbors disconnect the graph; too many erase local linearity

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Locally linear embedding: implementation

1. choose neighbors
2. regularize ill-conditioned local covariance
3. solve for weights
4. inspect neighborhood preservation

Check the result

Check trustworthiness and continuity together with the fitted transformation and data split.

Common mistake

too few neighbors disconnect the graph; too many erase local linearity

Do not trust Locally linear embedding before this check

Example

a pose manifold sampled by images

Explain the calculation and check trustworthiness and continuity, using a small example.

Common mistake

too few neighbors disconnect the graph; too many erase local linearity

What to do

Use LLE only when local patches are plausibly linear and sampling density is adequate.

Locally linear embedding: check your understanding

Question

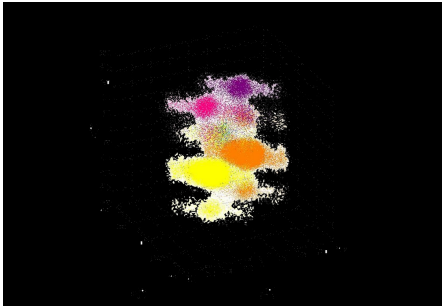
Can local reconstruction weights survive a change of coordinates?

Answer

The weights are invariant to local translation, rotation, and scaling under suitable neighborhoods.

Decision. Use LLE only when local patches are plausibly linear and sampling density is adequate.

What may be inferred from a visually separated two-dimensional embedding?



Question

What may be inferred from a visually separated two-dimensional embedding?

Example

visualizing learned digit representations

Main idea

t-SNE emphasizes conditional neighbor probabilities; UMAP emphasizes a fuzzy topological graph.

Distances between t-SNE clusters can mislead



t-SNE emphasizes local neighbors; visual distances between clusters need not match original distances.

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t-SNE, UMAP, and neighborhood preservation: model and variables

Model and assumptions

Neighbor-embedding methods match local fuzzy or probabilistic relations rather than preserving every global distance.

$$\mathrm{KL}(P\|Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

Symbol	Meaning	Status
p_{ij}	input-space affinity	controlled by perplexity or graph scale
q_{ij}	output-space affinity	heavy-tailed for t-SNE
KL	asymmetric mismatch	penalizes missing true neighbors strongly

t-SNE, UMAP, and neighborhood preservation: worked example

Quantity	Value	Meaning
perplexity	small	fine local groups
perplexity	large	broader context
cluster gap	visible	not a calibrated distance

What the result means

Use the map to generate hypotheses; use original-space or held-out evidence to confirm them.

t-SNE, UMAP, and neighborhood preservation: checking the result

Main idea

t-SNE emphasizes conditional neighbor probabilities; UMAP emphasizes a fuzzy topological graph.

What to check

trustworthiness at several k values

Limit

cluster size, gap, and orientation in the map can be optimization artifacts

t-SNE, UMAP, and neighborhood preservation: implementation

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```
from sklearn.manifold import TSNE
tsne = TSNE(n_components=2, perplexity=30, init='pca', learning_rate='auto', max_iter=500, random_state=7)
Z_tsne = tsne.fit_transform(X[:400])
print('t-SNE embedding:', Z_tsne.shape)
```


Evidence and failure check

Check trustworthiness at several k values with the fitted transformation and data split; reject the run if cluster size, gap, and orientation in the map can be optimization artifacts.

t-SNE, UMAP, and neighborhood preservation: limitations

Example

visualizing learned digit representations

Explain the calculation and check trustworthiness at several k values, using a small example.

Common mistake

cluster size, gap, and orientation in the map can be optimization artifacts

What to do

Use the map to generate hypotheses; use original-space or held-out evidence to confirm them.

t-SNE, UMAP, and neighborhood preservation: check your understanding

Question

What may be inferred from a visually separated two-dimensional embedding?

Answer

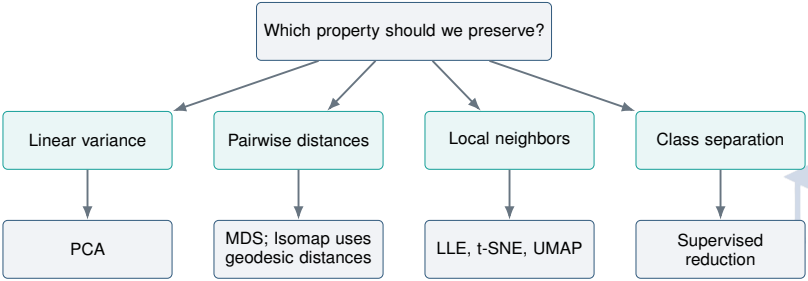
t-SNE emphasizes conditional neighbor probabilities; UMAP emphasizes a fuzzy topological graph.

Decision. Use the map to generate hypotheses; use original-space or held-out evidence to confirm them.

Compare methods by what they preserve

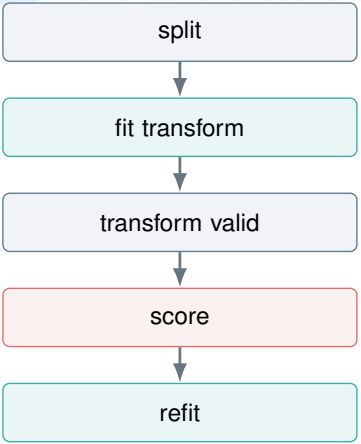
Method	Primary target	Main diagnostic
PCA	linear variance	variance and reconstruction
MDS	global distances	stress
Isomap	graph geodesics	residual variance
LLE	local reconstruction	trustworthiness
t-SNE / UMAP	local affinities	neighbor overlap across runs

Choose an embedding by what it should preserve



These are starting points. Check the distortion and the performance required by your task.

Keep reduction inside the validation boundary



No statistic crosses backward from validation to training.

1. split observations and groups
2. fit scaling and reduction on training folds
3. transform validation data
4. tune dimension and hyperparameters using validation evidence
5. refit once after the design is fixed

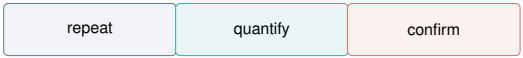
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Trustworthiness and continuity answer different questions

$T(k)$: false neighbors introduced, $C(k)$: true neighbors omitted

High T	High C	Interpretation
yes	yes	local neighborhoods agree
yes	no	some true neighbors split
no	yes	spurious neighbors appear

Stress-test the story told by a two-dimensional map



- repeat random seeds
- sweep neighborhood scales
- compare metrics and preprocessing
- measure neighbor overlap
- check conclusions in the original space
- separate exploration from confirmation

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Six takeaways to retain

1. define the preserved relation first
2. center before PCA and justify scaling
3. read PCA as variance and reconstruction
4. treat graph construction as part of the model
5. interpret local embeddings locally
6. validate stability before communicating clusters

Exit ticket: defend one embedding

Prompt

Choose one method for the case below. State the preservation target, one hyperparameter, one diagnostic, and one reason to reject the map.

Case

A hospital wants a two-dimensional view of 5,000 patient representations to explore possible subgroups, while retaining the original model for prediction.